



Sustainable Engineering 1: Economic and environmental issues in engineering

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Economic and Environmental Impact of Materials

- Concepts of sustainable engineering design
- Life cycle analysis
- Design for environment
- Materials processing considerations

Background information:
For these lectures read

Callister and Rethwisch 9e Chapter 23

Ashby, Shercliff and Cebon 3e

Environmental impact



- Aims and Learning objectives



Outline the process of life cycle analysis for assessing sustainability of materials and products



Explain the importance of considering environmental impact in the design of new materials and products



Review the effect of microstructure in processing of materials and present how these affect sustainable design



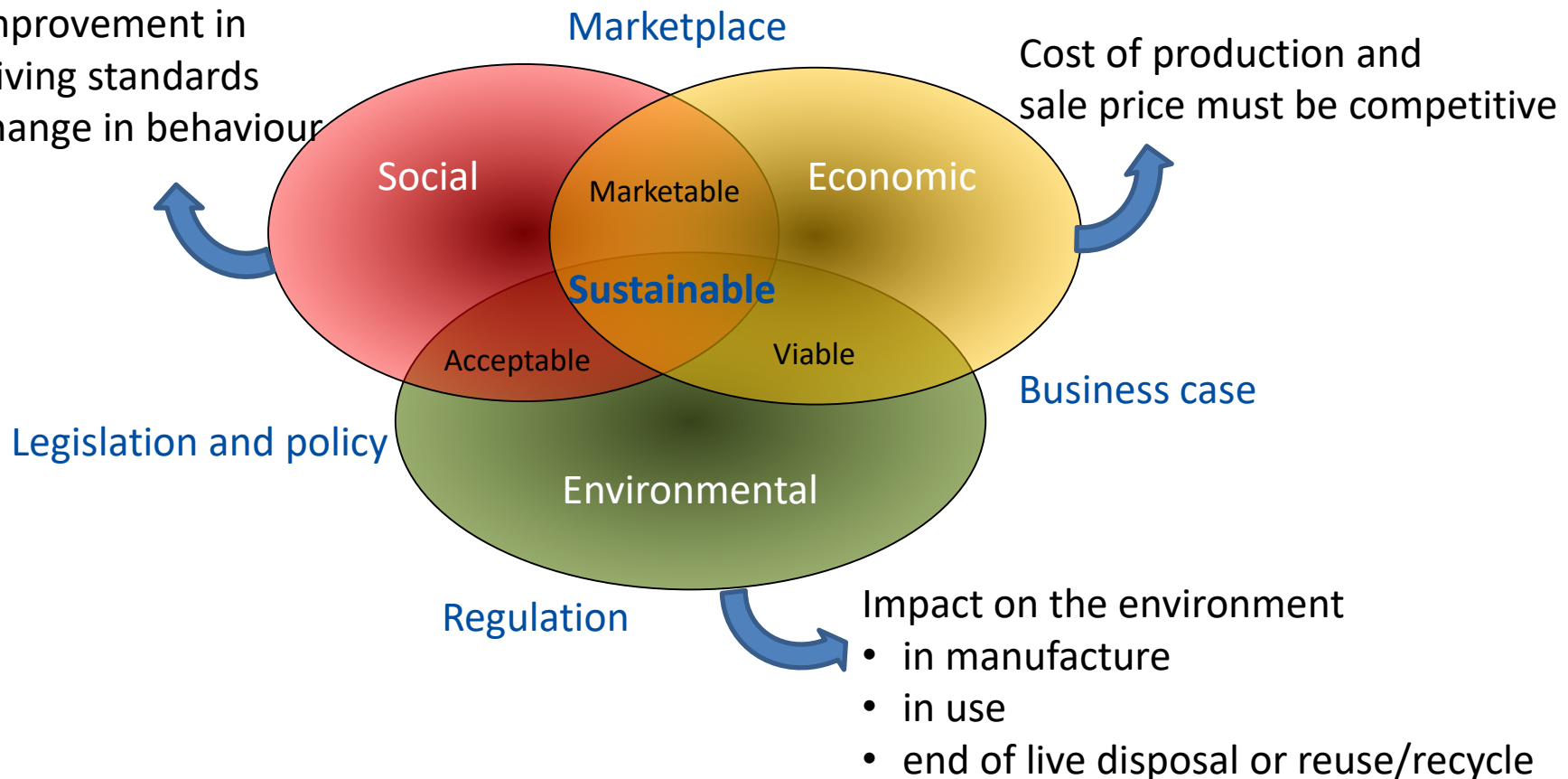
Link microstructure and properties to sustainable design

Sustainable engineering

Sustainable engineering requires consideration of 3 factors :

Impact on society

- improvement in living standards
- change in behaviour



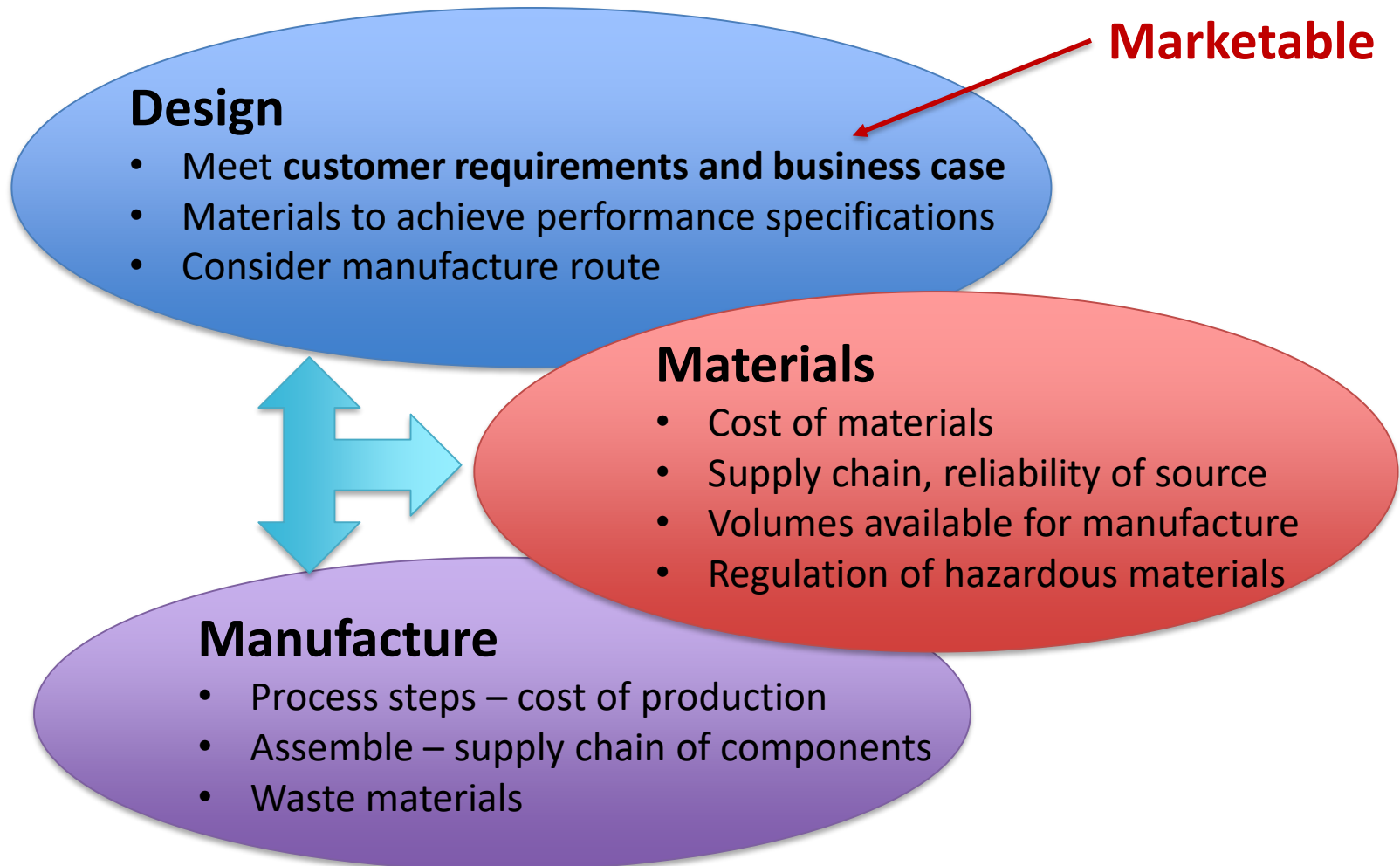
Social impact of technology

- Advances in technology enable improvements in quality of life, for example:
 - More efficient use of energy & resources
 - Improved healthcare
 - Improved access to information
 - Improved security

- Technology can change the way we interact in society, for example:
 - Communications
 - Transport
 - Movement of goods and information

Economics of engineering

Engineering design includes these inter-related aspects:



Economics of engineering

Examples of materials costs - metals (price /kg relative to steel)

Low carbon steel	1.0
Alloy steel	2.8
Stainless steel	7.3
Cast iron	2.0
Aluminium	2.1
Aluminium alloy (7075 T6)	10
Copper	7.4
Nickel	24
Inconel 624	55
Titanium	85
Titanium 6-4	95
Tungsten	175
Gold	30,000

Economics of engineering

Examples of materials costs – ceramics and glasses (price / kg relative to steel)

Concrete	0.04
Alumina	1.8
Zirconia	97
Silicon Carbide	250
Silicon nitride	194
Soda-lime glass	0.7
Borosilicate glass	7.9
Fused silica	1100
Silicon	5500
Gallium Arnside	5000
Diamond	6000
Graphite	65

Economics of engineering

Examples of materials costs – polymers and composites (price / kg relative to steel)

Polyethelene	1.3
UHMWPE	2.1
Polystyrene	1.4
PA66 (nylon)	2.8
PMMA	3.1
Epoxy resin	1.9
SBR rubber	1.3
Silicone rubber	9.5
Aramid fibre	38.8
Carbon fibre	193
Carbon-fibre-epoxy composite	330

Elements considered critical for technology



Some elements are considered critical for advanced technologies, such as high performance materials, energy production, electronic or functional materials

Critical elements

(US Government sources)

1 H																	2 He										
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne										
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar										
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr										
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe										
55 Cs	56 Ba											72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	
87 Fr	88 Ra											104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg								
			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu										
			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr										

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By laws, such as EU REACH regulations

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Stages of material manufacture

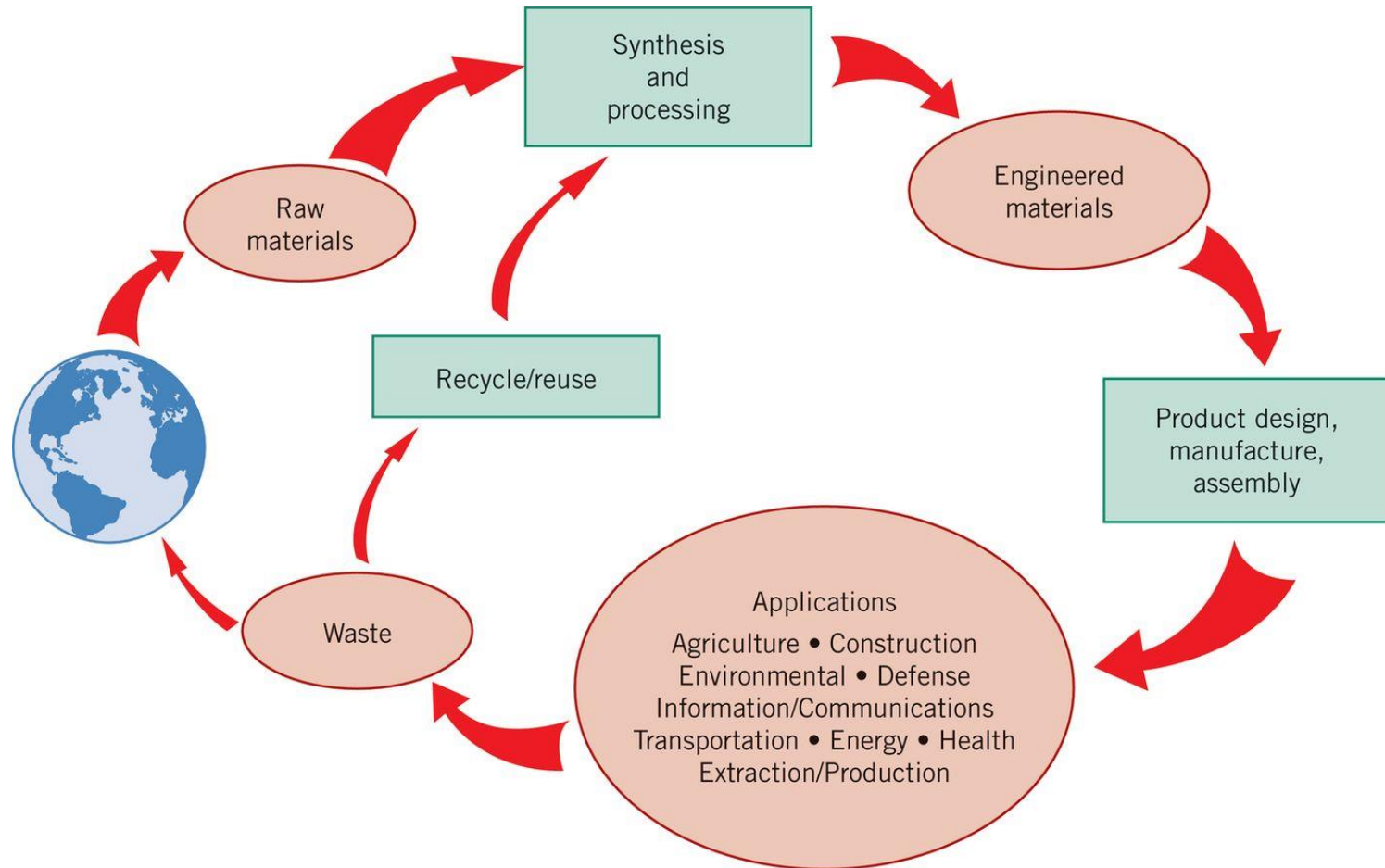


Fig. 23.1, Callister & Rethwisch 9e..

Stages of material manufacture

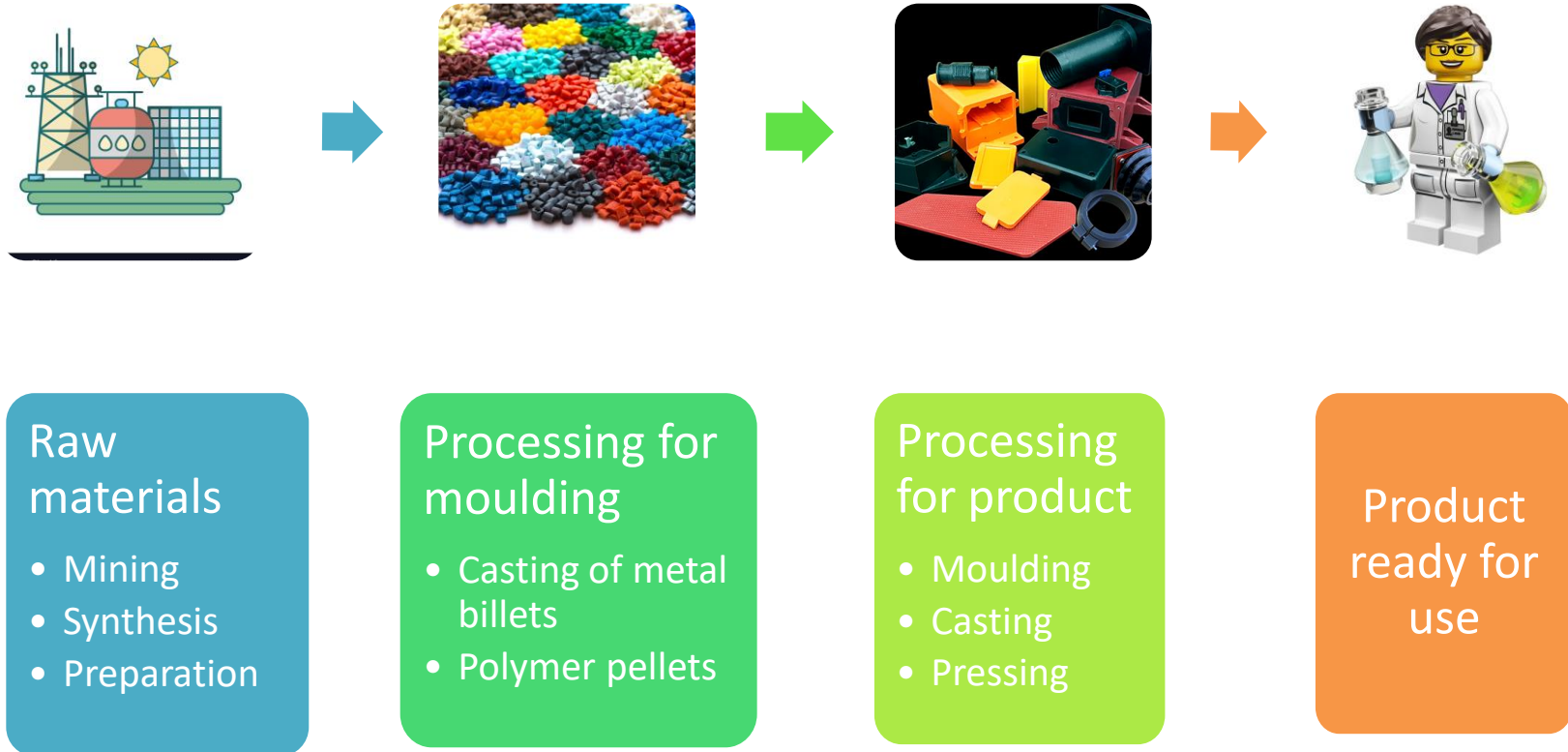
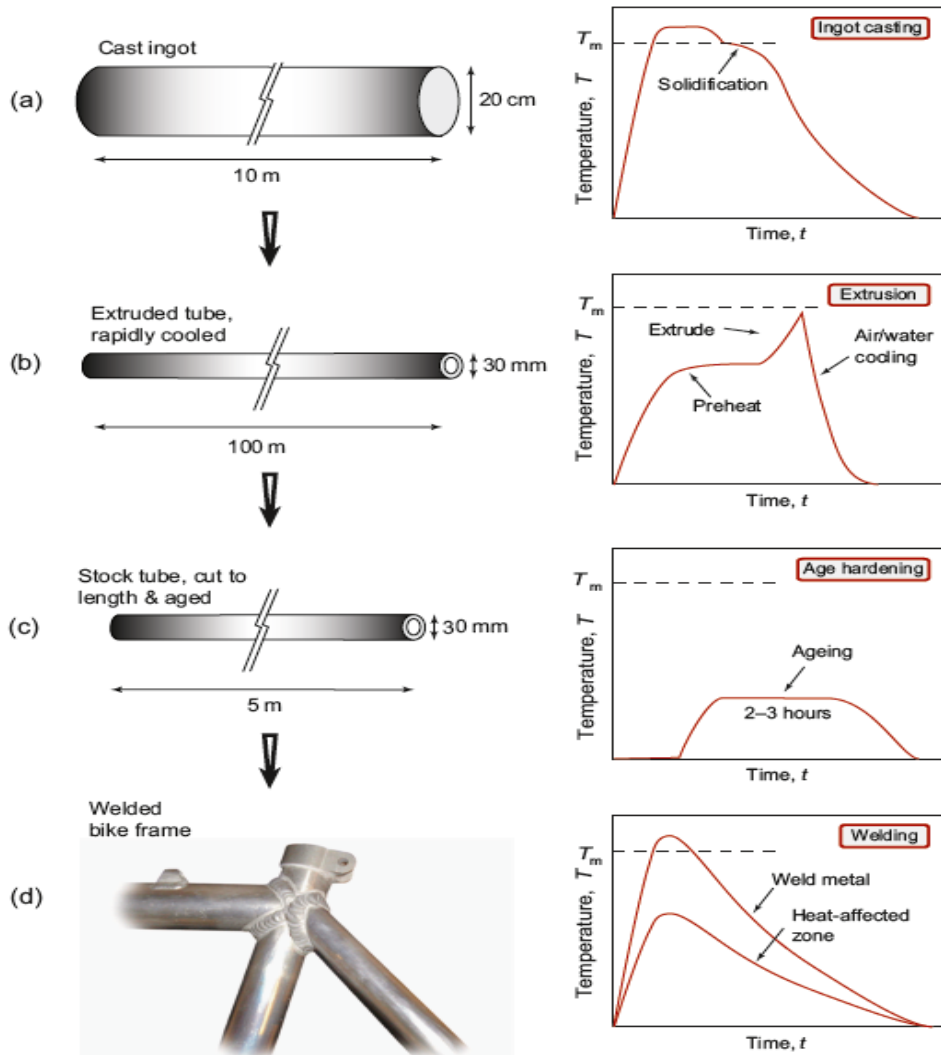


Fig. 21.10, Callister & Rethwisch 9e..

Processing and properties

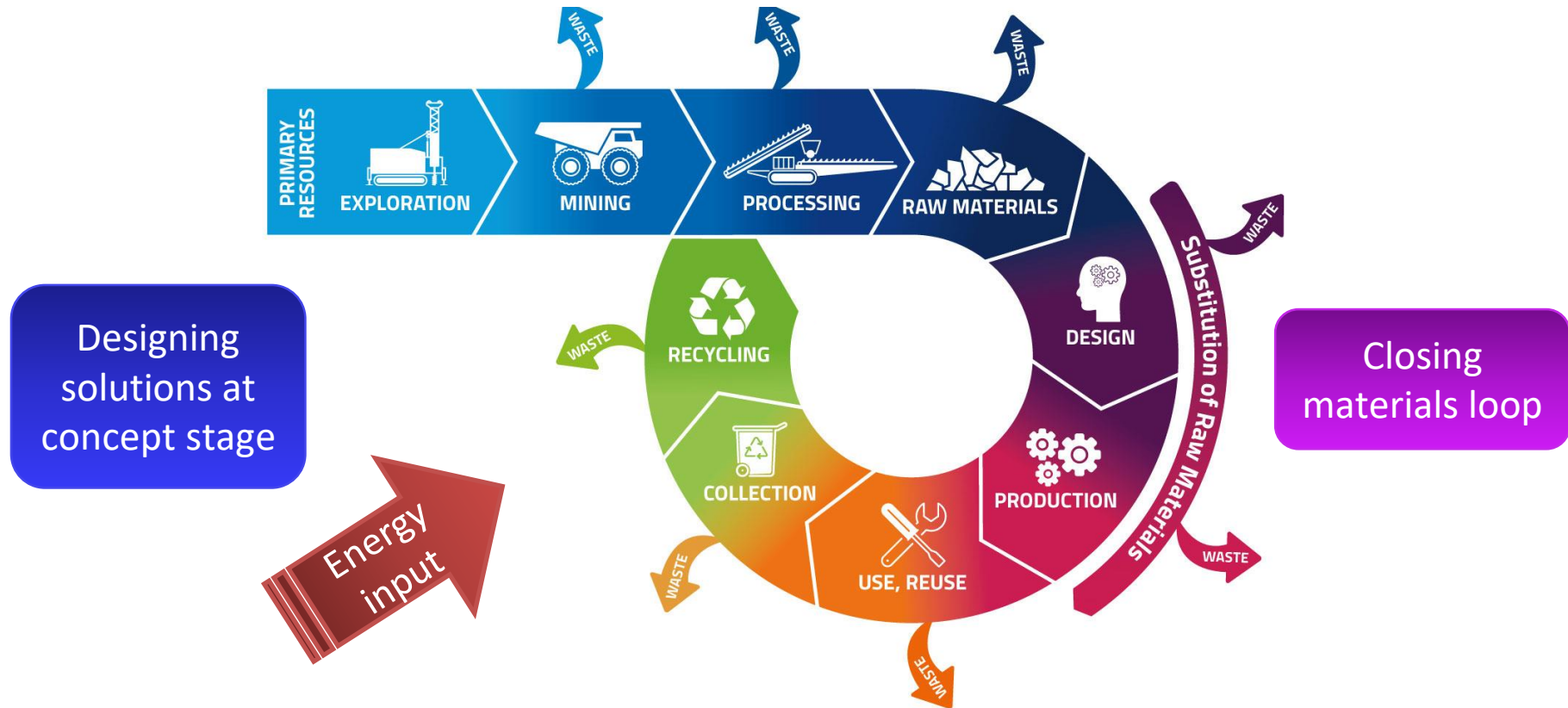
Process history of an aluminium bicycle frame



- Processing materials involves several steps
- Each process has specific requirements (e.g. temperature, pressure, strength, ductility)
- Processing of materials affects properties due to changes in the microstructure
- Designing for specific properties involves using processes that result in the desired properties

Environmental impact

Linking environmental sustainability to the production of materials and products



Environmental impact

Processing does not consider environmental impact

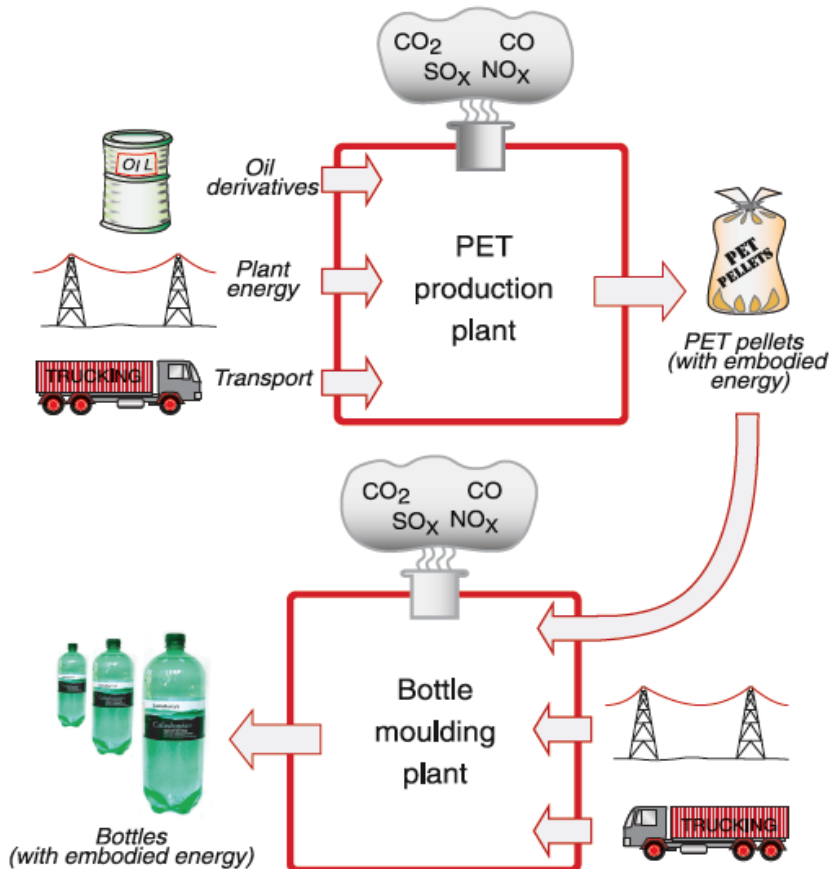


Figure 20.4

- Energy required for extraction, preparation, processing of raw feedstock
- Emissions released from transformation activities
- Energy and emissions used during storage, transportation and commercialization of finished goods to consumers

Sustainable development



Can be defined as:

*“Sustainable development is :
development that meets the needs of
the present without compromising the
ability of future generations to meet
their own needs”*

Report of the Brundtland commission of the UN, 1987